

BAVISH KULUR

Research Interests: Reinforcement Learning, Deep Learning, Continual Learning
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Education

University of Alberta 2025 - Present
MSc (Thesis) - Computing Science GPA: 4.0/4.0
Advisor: Prof. Marlos C. Machado
Courses: Reinforcement Learning I & II, Foundations of Reasoning in LLMs, ML II

Indian Institute of Technology Bombay 2017 - 2022
Dual Degree - BTech in Mechanical Engineering and MTech in AI and Data Science GPA: 8.84/10
Minor in Computer Science and Engineering
Relevant Courses: Advanced ML, Foundations of Intelligent and Learning Agents, Learning over Graphs, Operating Systems, Data Structures and Algorithms

Publications

[AAMAS] Richa Verma, **Bavish Kulur**, Sanjay Chawla, Balaraman Ravindran.
PREFINE: Preference-Based Implicit Reward and Cost Fine-Tuning for Safety Alignment
International Conference on Autonomous Agents and Multiagent Systems 2026 [Paper Link]

Experience

Post-Baccalaureate Research Fellow Robert Bosch Centre for Data Science and AI Nov 23 - Jun 25
Indian Institute of Technology Madras, Advisor: Prof. Balaraman Ravindran

Product Engineer (SDE 1) Sprinklr Ltd. Gurugram, India Jul 22 - Feb23

Software Engineering Intern Arturo Technologies, Bangalore, India Mar - Jun 19

Technical Skills

Machine Learning: Python, PyTorch, JAX

Software Development: C++, Java, Spring, Javascript, ReactJS, MongoDB, SQL, Docker, Git, Linux

Key Projects

Exploration in Continual RL Sept 25 - Present
Advisor: Prof. Marlos Machado (UAlberta)
Working on RL algorithms for exploration in reset-free environments.

GARE: Greedy Actor Regularized Expert (Course Project) Sept - Dec 25
Advisor: Prof. Martha White (UAlberta)
Proposed a value-based RL algorithm for continuous action spaces, where the maximal action for TD update is found using gradient ascent starting from a proposal action given by a learned function. Showed that it outperforms the baseline algorithms GreedyAC and SAC.

Safe Reinforcement Learning using Direct Preference Optimization 2024 - 25
 Advisors: Prof. Balaraman Ravindran (IIT Madras), Dr. Sanjay Chawla (QCRI, Qatar)
Offline post-hoc fine-tuning of RL policy to make it safe and aligned to human preferences.

Fairness Evaluation of Distilled LLMs 2024 - 25
 Advisors: Prof. Balaraman Ravindran (IIT Madras) and Prof. Karthik Sheshadri (NIT AP)
Task-specific models can be built through knowledge distillation of LLMs. Prior work in knowledge distillation methods only considers computation time and accuracy metrics. In this work, we investigated the effect of knowledge distillation on fairness metrics.

Microstructure CT Image Segmentation (Thesis project) 2021 - 22
 Advisor: Prof. Shyamprasad Karagadde (IIT Bombay)
Trained U-Net based image segmentation models for segmenting microstructures in real-world CT images of materials (aluminum alloys) and performed analysis such as computation of porosity, etc. Integrated the trained model into ImageJ software.

Parallelizing Stochastic FEM Analysis (Course project) Jan - Apr 2021
 Advisor: Prof. S. Gopalakrishnan (IIT Bombay)
Parallelized the Response Surface algorithm for Stochastic Finite Element Analysis. Reduced the computation time by 35% by implementing CUDA kernel functions to use GPU and multi-threading using OpenMP.

Teaching Assistantships

CMPUT356 Reinforcement Learning, University of Alberta	Sept - Dec 2025
Introduction to Machine Learning, NPTEL (MOOC)	Jan - Apr 2025
CS101 Computer Programming and Utilization, IIT Bombay	Jan - Apr 2019

Achievements

Selected to attend the Reinforcement Learning workshop at Indian Institute of Science	2024
Ranked 4 (out of 20) in AI and Data Science batch at IIT Bombay	2022
Received Student Innovation grant from ARTPARK, Indian Institute of Science	2021
Selected for a semester exchange at National University of Singapore (NUS)	2020
Received Kishore Vaigyanik Protsahan Yojana (KVPY) award from Department of Science and Technology, Government of India	2017
Cleared Regional Mathematical Olympiad and attended Indian National Mathematics Olympiad training camp at Indian Statistical Institute, Bangalore	2015

Activities

Convener, Web and Coding Club, IIT Bombay	2018 - 19
Developer and Founding Member, InstiApp, Developers Community, IIT Bombay	2018 - 19